Vol 3 Chapter 6 — Superheavy Island of Stability

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Chapter 6 — Superheavy Island of Stability

The superheavy island of stability — a predicted region of relatively long-lived superheavy nuclei near $Z \approx 114$ ($N \approx 184$) — has been a target of nuclear-experiment programs for fifty years. Predictions of the island's exact location and the stabilities of its members vary across standard shell-model calculations.

BST derives the substrate's predicted island center from BST primary integers.

6.1 The substrate-derived island center

Per substrate spin-orbit coupling $\kappa_{ls} = 6/5$ extended to the superheavy regime, the next doubly-magic shell closure beyond Pb-208 is predicted at:

$$Z = 114, \quad N = 184,$$

— flerovium-298. Substrate-derived half-life predictions for the island center: hundreds of years to thousands of years range, consistent with the standard superheavy-island predictions but with structural derivation rather than empirical fit.

6.2 Experimental status

Elements 113 (nihonium), 114 (flerovium), 115 (moscovium), 116 (livermorium), 117 (tennessine), 118 (oganesson) have all been synthesized at FRIB / JINR / GSI. Their half-lives range from microseconds to seconds — consistent with the substrate's predicted approach to the island center at $Z = 114, N = 184$.

The substrate framework's prediction is that flerovium-298 specifically, when synthesized, will have anomalous stability compared to its neighbors — multi-year experimental falsifier.

6.3 What comes next

Chapter 7 — Atomic Orbital Sequence — derives the atomic shell structure $(2l + 1) = 1, N_c, n_C, g$ from substrate primaries.

Where to look this up: Superheavy-Z predictions: Task #111. For standard superheavy physics: Oganessian 2007 review.